HIT137

Group Assignment 3 (30% Mark)

You are required to create a GitHub repository and add all your group mates to it (make sure to keep it public, not private). You should do this before you start the assignment.

All the answers and contributions should be recorded in GitHub till you submit the assignment.

Submission Guidelines:

- **Include your GitHub Repository link in a text file “github_link.txt”.**
- **Zip all the programming files, outputs, and “github_link.txt”. Upload them to Learline.**

Overview

You will develop a desktop application that demonstrates your understanding of OOP, GUI development using Tkinter, and image processing using OpenCV.

Build a desktop application where two nearly identical images are displayed side by side. One image is the original; the other is a programmatically altered copy containing hidden differences. The player clicks on the modified image to locate the differences, and the application validates each click against the known difference regions.

Functional Requirements

1. Object-Oriented Programming

Your application must be structured using proper OOP principles:

- The codebase must be organised into at least three classes
- Demonstrate encapsulation, constructor, methods, class interaction, inheritance and polymorphism

2. Image Processing with OpenCV

When an image is loaded, the application should create an exact clone of the original and programmatically introduce exactly 5 differences into the clone at random locations. The 5 difference regions must not overlap one another. All 5 differences are generated at once. Each time an image is loaded, the type and position of each difference should be chosen randomly.

The program should support at least three types of alteration. Feel free to choose the alteration methods. Example of alteration:

- Colour Shift. A rectangular region of the cloned image has its colour properties shifted so that it is visually different from the original. The change should be noticeable upon careful inspection but not glaringly obvious.

All image manipulation must be performed using OpenCV.

3. Tkinter GUI

Image Loading and Display

A button allows the user to choose an image from disk, support for common image formats (JPG, PNG, BMP).

The original image and the modified image must be displayed side by side. The original image is shown on the left for visual reference only. The modified image is shown on the right and is the only image that responds to player clicks.

Finding Differences

A visible section of the interface displays the number of differences still unfound for the current image (e.g., "Remaining: 3"). This is cumulative score across multiple images.

When the player clicks on the modified image, the application check whether the click falls within a reasonable proximity to any unfound difference region.

If the click matches an unfound difference, that difference is marked as found and a red circle is drawn around the difference region on both the original and the modified image. When all 5 differences in the current image have been found, the player should be notified, for example, via a pop-up dialog or an on-screen message. The player can then load another image to continue playing.

Mistakes

If the click does not match any unfound difference, it counts as a mistake. The current number of mistakes should be displayed on the interface.

The player is allowed a maximum of 3 mistakes per image. When the player reaches 3 mistakes, no further guesses are allowed. A clear visual prompt should be displayed to inform the player that they have made too many incorrect guesses and number of differences have been found. The player can then load a new image to restart.

Reveal

A button should be available to reveal all unfound differences. When pressed, all currently unfound differences are visually marked with a blue circle on both images. After revealing, the player can then load a new image to restart.